材 料 汇 总 表

材料	材质	规格	段号 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	330-KC22D-DJ (0°-40°)呼称高(m) 15.0	18.0	21.0	24.0	27.0	30.0	330-KC22D-DJ (40°-90°)呼称高(m) 15.0	18.0	21.0	24.0	27.0	30.0
角钢	Q420	L200X16																				2396.7	1188.9	1803.1					2396.7	2992.0					2396.7	2992.0
		L180X16														814.8	1347.3	1880.0	797.2	1613.7	797.2				814.8	1347.3	1880.0	2410.9	797.2	797.2	814.8	1347.3	1880.0	2410.9	797.2	797.2
		L180X14													1218.3							118.2		118.2	1218.3	1218.3	1218.3	1218.3	1336.5	1336.5	1218.3	1218.3	1218.3	1218.3	1336.5	1336.5
		L160X14																		104.7		104.7	104.7					104.7	104.7	104.7				104.7	104.7	104.7
		L160X12													88.8	90.5	90.5	90.5	90.5		90.5				179.3	179.3	179.3	179.3	179.3	179.3	179.3	179.3	179.3	179.3	179.3	179.3
		L140X12								960.4				572.8											572.8	572.8	572.8	572.8	572.8	572.8	1533.2	1533.2	1533.2	1533.2	1533.2	1533.2
		L125X10				578.4																			578.4	578.4	578.4	578.4	578.4	578.4						
		小 计				578.4				960.4				572.8	1307.1	905.3	1437.8	1970.5	887.7	1718.4	887.7	2619.6	1293.6	1921.3	3363.6	3896.1	4428.8	5064.4	5965.6	6560.9	3745.6	4278.1	4810.8	5446.4	6347.6	6942.9
	Q345	L140X10										422.8	422.8	336.6											759.4	759.4	759.4	759.4	759.4	759.4	759.4	759.4	759.4	759.4	759.4	759.4
		L125X10			505.6				503.4	132.2				122.2											627.8	627.8	627.8	627.8	627.8	627.8	757.8	757.8	757.8	757.8	757.8	757.8
		L125X8												39.1	303.4										342.5	342.5	342.5	342.5	342.5	342.5	342.5	342.5	342.5	342.5	342.5	342.5
		L110X8													717.2										717.2	717.2	717.2	717.2	717.2	717.2	717.2	717.2	717.2	717.2	717.2	717.2
		L100X8			25.4	28.0			24.5					263.4	715.8										1032.6	1032.6	1032.6	1032.6	1032.6	1032.6	1003.7	1003.7	1003.7	1003.7	1003.7	1003.7
		L100X7									175.2			180.2											355.4	355.4	355.4	355.4	355.4	355.4	355.4	355.4	355.4	355.4	355.4	355.4
		L90X7	31.2				31.2	131.6		44.4		21.2	21.2	145.4											197.8	197.8	197.8	197.8	197.8	197.8	373.8	373.8	373.8	373.8	373.8	373.8
		L80X7				98.6				79.4	25.1	186.4	188.8												211.5	211.5	211.5	211.5	211.5	211.5	391.9	391.9	391.9	391.9	391.9	391.9
		L80X6	85.4	71.0						32.4	96.4	186.8	186.8	105.8											545.4	545.4	545.4	545.4	545.4	545.4	421.4	421.4	421.4	421.4	421.4	421.4
		L75X6						90.2		31.1																					121.3	121.3	121.3	121.3	121.3	121.3
		L75X5	50.0		15.6	15.8			23.3	18.6		87.6	87.6										133.8		169.0	169.0	169.0	169.0	169.0	302.8	129.5	129.5	129.5	129.5	129.5	263.3
		L70X6				43.2	55.2			59.2		168.2	169.1												211.4	211.4	211.4	211.4	211.4	211.4	283.5	283.5	283.5	283.5	283.5	283.5
		L70X5		51.2	24.1	113.8			19.0															164.4	189.1	189.1	189.1	189.1	189.1	353.5	19.0	19.0	19.0	19.0	19.0	183.4
		L63X5	44.4	7.3	80.4	30.0	59.2	7.2	80.2	4.4		10.1	11.2												172.2	172.2	172.2	172.2	172.2	172.2	162.2	162.2	162.2	162.2	162.2	162.2
		小 计	211.0	129.5	651.1	230.8	244.2	229.0	650.4	401.7	296.7	1083.1	1087.5	1192.7	1736.4								133.8	164.4	5531.3	5531.3	5531.3	5531.3	5531.3	5829.5	5838.6	5838.6	5838.6	5838.6	5838.6	6136.8
	Q235	L125X8															951.2	1504.4		1162.6		1773.0	1237.6			951.2	1504.4	1162.6	1773.0	1237.6		951.2	1504.4	1162.6	1773.0	1237.6
		L110X8														544.0			858.8		882.8			1068.0	544.0			858.8	882.8	1950.8	544.0			858.8	882.8	1950.8
		L100X7																				337.0		379.5					337.0	379.5					337.0	379.5
		L90X7												61.7											61.7	61.7	61.7	61.7	61.7	61.7	61.7	61.7	61.7	61.7	61.7	61.7
		L80X7														199.8	199.8	223.7		242.8					199.8	199.8	223.7	242.8			199.8	199.8	223.7	242.8		
		L80X6								114.1														180.6						180.6	114.1	114.1	114.1	114.1	114.1	294.7
		L75X6								103.2														157.8						157.8	103.2	103.2	103.2	103.2	103.2	261.0
		L70X6																142.1				397.6		256.5					397.6	256.5			142.1		397.6	256.5
		L70X5				43.6					51.8					203.4	345.8	455.3	101.0	550.2	101.2	507.3	232.0	265.3	298.8	441.2	550.7	746.6	703.9	693.9	255.2	397.6	507.1	703.0	660.3	650.3
		L63X5	75.6		123.9	44.7	75.6		124.0	15.1	234.2	30.0	30.0	38.9	123.8	135.6	282.2	286.2	133.1	390.2	133.3	368.9	158.2	215.7	806.7	953.3	957.3	1194.4	1173.3	1178.3	777.2	923.8	927.8	1164.9	1143.8	1148.8
		L56X5			5.4				6.4	14.0				27.9	16.5									171.2	49.8	49.8	49.8	49.8	49.8	221.0	64.8	64.8	64.8	64.8	64.8	236.0
		L56X4			19.0	19.2	22.2	13.8	18.4	90.4	51.3							104.4		255.6		316.9		236.7	89.5	89.5	193.9	345.1	406.4	326.2	196.1	196.1	300.5	451.7	513.0	432.8
		L50X5													281.1				69.1		69.0				281.1	281.1	281.1	350.2	350.1	350.1	281.1	281.1	281.1	350.2	350.1	350.1
		L50X4	19.3	11.3	20.2	21.8	6.5	11.1	20.0	56.8		14.5	14.5			108.6	108.6	28.3		113.5		21.2			195.7	195.7	115.4	200.6	108.3	87.1	217.5	217.5	137.2	222.4	130.1	108.9
		L45X4	65.8	51.7	3.4	3.5	59.9	74.7	3.3	9.5	7.9			15.1	35.1		97.3	228.2		17.4				98.7	182.5	279.8	410.7	199.9	182.5	281.2	205.5	302.8	433.7	222.9	205.5	304.2
		L40X4	56.6	7.5	6.6	20.2	61.8		6.2	36.1				25.4	18.9	99.5	104.7	9.0		59.4		158.3		10.4	234.7	239.9	144.2	194.6	293.5	145.6	247.9	253.1	157.4	207.8	306.7	158.8
		L40X3	65.2	35.3	25.2	16.9	55.5	48.7	24.6	30.9				83.9		69.6	54.8	60.6		108.4		99.3		160.0	296.1	281.3	287.1	334.9	325.8	386.5	313.2	298.4	304.2	352.0	342.9	403.6
		小 计	282.5	105.8	203.7	169.9	281.5	148.3	202.9	470.1	345.2	44.5	44.5	252.9	475.4	1360.5	2144.4	3042.2	1162.0	2900.1	1186.3	3979.5	1627.8	3200.4	3240.4	4024.3	4922.1	5942.0	7045.7	7894.4	3581.3	4365.2	5263.0	6282.9	7386.6	8235.3
钢板	Q420	-10														75.0	75.0	75.0	75.0		75.0				75.0	75.0	75.0	75.0	75.0	75.0	75.0	75.0	75.0	75.0	75.0	75.0
		-12				242.2									255.0					89.9		194.5	90.0	104.5	497.2	497.2	497.2	587.1	691.7	691.7	255.0	255.0	255.0	344.9	449.5	449.5
		-14								344.0																					344.0	344.0	344.0	344.0	344.0	344.0
		小 计				242.2				344.0					255.0	75.0	75.0	75.0	75.0	89.9	75.0	194.5	90.0	104.5	572.2	572.2	572.2	662.1	766.7	766.7	674.0	674.0	674.0	763.9	868.5	868.5
	Q345	-6			20.8	31.2			19.8	5.3		28.7	28.7												80.7	80.7	80.7	80.7	80.7	80.7	53.8	53.8	53.8	53.8	53.8	53.8
		-8	38.3	14.2	26.0	28.2	55.7		26.6	39.2		110.1	110.1	44.9											261.7	261.7	261.7	261.7	261.7	261.7	276.5	276.5	276.5	276.5	276.5	276.5
		-10	68.6	68.6	18.3	20.1	68.6	85.2	18.8	50.0		42.2	41.0	126.3	188.7	72.3	75.4	93.2	81.4		36.1				605.1	608.2	626.0	614.2	568.9	568.9	650.9	654.0	671.8	660.0	614.7	614.7
		-12			240.2				231.2	30.2															240.2	240.2	240.2	240.2	240.2	240.2	261.4	261.4	261.4	261.4	261.4	261.4
		-16	32.4	27.6			74.0	28.2								18.8	48.2	52.8		55.7		52.8		48.2	78.8	108.2	112.8	115.7	112.8	108.2	121.0	150.4	155.0	157.9	155.0	150.4
		-18														362.0	362.0	362.0		370.1		365.4		365.4	362.0	362.0	362.0	370.1	365.4	365.4	362.0	362.0	362.0	370.1	365.4	365.4
		-22			149.2	149.2			150.0	149.2		117.2	117.2												415.6	415.6	415.6	415.6	415.6	415.6	416.4	416.4	416.4	416.4	416.4	416.4
		-56																		633.0								633.0						633.0		
		-58														655.6	655.6	655.6				655.6		655.6	655.6	655.6	655.6		655.6	655.6	655.6	655.6	655.6		655.6	655.6
		小 计	139.3	110.4	454.5	228.7	198.3	113.4	446.4	273.9		298.2	297.0	171.2	188.7	1108.7	1141.2	1163.6	81.4	1058.8	36.1	1073.8		1069.2	2699.7	2732.2	2754.6	2731.2	2700.9	2696.3	2797.6	2830.1	2852.5	2829.1	2798.8	2794.2
	Q235	-2										1.8	1.8	3.7		6.5	6.5	6.5	6.5		6.0				12.0	12.0	12.0	12.0	11.5	11.5	12.0	12.0	12.0	12.0	11.5	11.5
		-5	38.4	6.9	6.6	6.2	40.4	10.4	6.6	11.7	19.6				9.4	18.8	17.1	17.1		10.4		20.2		19.0	105.9	104.2	104.2	97.5	107.3	106.1	116.9	115.2	115.2	108.5	118.3	117.1
		-6			27.8	29.2			26.8	20.8	56.2	10.5	10.5	17.2	15.5	34.8	56.9	59.2		69.8		83.2		64.2	191.2	213.3	215.6	226.2	239.6	220.6	181.8	203.9	206.2	216.8	230.2	211.2
		-8	77.8				44.1			38.8				0.2				38.2				39.9		61.4	78.0	78.0	116.2	78.0	117.9	139.4	83.1	83.1	121.3	83.1	123.0	144.5
		-10	0.6	0.6	2.8	3.4	0.2		3.1		0.4	4.5	4.5	0.6		58.3	59.2	133.4		134.2		117.9		47.0	71.2	72.1	146.3	147.1	130.8	59.9	67.1	68.0	142.2	143.0	126.7	55.8
		-12	0.5				1.2	1.0		3.1		2.0	2.0	0.7	25.8								1.4		29.0	29.0	29.0	29.0	29.0	30.4	33.8	33.8	33.8	33.8	33.8	35.2
		-14		0.3				0.3				0.8	0.8		9.5										10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6	10.6
		-16			1.7	1.2			1.2						2.7			1.8	3.6		3.6				5.6	5.6	7.4	9.2	9.2	9.2	3.9	3.9	5.7	7.5	7.5	7.5
		-18			0.5	0.5			0.5			2.0	2.0						2.0						3.0	3.0	3.0	5.0	3.0	3.0	2.5	2.5	2.5	4.5	2.5	2.5
		-20								2.3																					2.3	2.3	2.3	2.3	2.3	2.3
		-22								0.4																					0.4	0.4	0.4	0.4	0.4	0.4
		-26													1.8						2.9				1.8	1.8	1.8	1.8	4.7	4.7	1.8	1.8	1.8	1.8	4.7	4.7
		小 计	117.3	7.8	39.4	40.5	85.9	11.7	38.2	77.1	76.2	21.6	21.6	22.4	64.7	118.4	139.7	256.2	12.1	214.4	12.5	261.2	1.4	191.6	508.3	529.6	646.1	616.4	663.6	595.4	516.2	537.5	654.0	624.3	671.5	603.3
钢管	Q345	Ø60/Ø38			1.8	1.8			1.8	1.8		1.8	1.8												5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4	5.4
		Ø30/Ø20	0.1	0.1			0.2	0.1																	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3
		小 计	0.1	0.1	1.8	1.8	0.2	0.1	1.8	1.8		1.8	1.8												5.6	5.6	5.6	5.6	5.6	5.6	5.7	5.7	5.7	5.7	5.7	5.7
螺栓	6.8级	M16X40	31.0	7.4	6.6	5.0	29.6	9.8	6.1	9.2	7.5	0.6	0.6	3.7	0.6	18.5	23.1	23.1		26.5		28.8		28.8	80.9	85.5	85.5	88.9	91.2	91.2	85.6	90.2	90.2	93.6	95.9	95.9
		M16X50	6.9	5.4	3.0	3.4	7.5	6.2	3.4	6.2	1.1	0.2	0.2	7.5	18.6	10.3	16.0	14.7	2.6	17.9	0.6	18.6		17.3	56.4	62.1	60.8	66.6	65.3	64.0	61.0	66.7	65.4	71.2	69.9	68.6
		M16X60			0.2	0.2			0.2	1.8				0.4	1.4										2.2	2.2	2.2	2.2	2.2	2.2	3.8	3.8	3.8	3.8	3.8	3.8
	6.8级(带双帽)	M16X60	0.8				0.8																		0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8
		M16X70	1.6	0.8	0.8	0.8	2.4	0.8	0.8																4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0	4.0
		M16X80			1.8	0.9			1.8	2.6															2.7	2.7	2.7	2.7	2.7	2.7	4.4	4.4	4.4	4.4	4.4	4.4
		小 计	40.3	13.6	12.4	10.3	40.3	16.8	12.3	19.8	8.6	0.8	0.8	11.6	20.6	28.8	39.1	37.8	2.6	44.4	0.6	47.4		46.1	147.0	157.3	156.0	165.2	166.2	164.9	159.6	169.9	168.6	177.8	178.8	177.5
	6.8级	M20X45	32.4	7.0	23.2	23.8	28.1	1.1	22.7	20.8	36.2	31.9	31.9	16.1	10.8	13.0	21.6	28.1	6.4	25.9	1.6	36.7	8.6	25.9	194.4	203.0	209.5	213.7	219.7	217.5	180.7	189.3	195.8	200.0	206.0	203.8
		M20X55	16.5	14.8	56.7	52.6	18.3	21.9	51.1	26.0	2.4	57.3	57.3	31.0	48.4	33.1	44.9	57.9	4.6	47.2	1.2	59.1	11.8	44.9	312.8	324.6	337.6	331.5	340.0	337.6	289.5	301.3	314.3	308.2	316.7	314.3
		M20X65			9.9	10.6			9.9	38.5		6.4	6.4	32.1	16.7	17.0	21.2	21.8	11.3	23.1	2.5	10.3	6.4	20.5	92.7	96.9	97.5	110.1	88.5	105.1	120.6	124.8	125.4	138.0	116.4	133.0
		M20X75								9.7					45.6				2.7		1.0				45.6	45.6	45.6	48.3	46.6	46.6	55.3	55.3	55.3	58.0	56.3	56.3
	6.8级(带双帽)	M20X60	5.8				5.8																		5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8	5.8
		M20X70	8.5	5.4	3.1	3.1	15.5	5.4	3.1	3.1															20.1	20.1	20.1	20.1	20.1	20.1	27.1	27.1	27.1	27.1	27.1	27.1
		M20X80			18.1	19.8			18.1	18.1		14.8	14.8												52.7	52.7	52.7	52.7	52.7	52.7	51.0	51.0	51.0	51.0	51.0	51.0
		小 计	63.2	27.2	111.0	109.9	67.7	28.4	104.9	116.2	38.6	110.4	110.4	79.2	121.5	63.1	87.7	107.8	25.0	96.2	6.3	106.1	26.8	91.3	724.1	748.7	768.8	782.2	773.4	785.4	730.0	754.6	774.7	788.1	779.3	791.3
	8.8级	M24X75														85.9	85.9	85.9	51.2	34.4	12.8	22.1		34.4	85.9	85.9	85.9	85.6	34.9	47.2	85.9	85.9	85.9	85.6	34.9	47.2
		M24X85																		55.1		71.2	55.1	55.1				55.1	71.2	110.2				55.1	71.2	110.2
		小 计														85.9	85.9	85.9	51.2	89.5	12.8	93.3	55.1	89.5	85.9	85.9	85.9	140.7	106.1	157.4	85.9	85.9	85.9	140.7	106.1	157.4
		螺栓合计	103.5	40.8	123.4	120.2	108.0	45.2	117.2	136.0	47.2	111.2	111.2	90.8	142.1	177.8	212.7	231.5	78.8	230.1	19.7	246.8	81.9	226.9	957.0	991.9	1010.7	1088.1	1045.7	1107.7	975.5	1010.4	1029.2	1106.6	1064.2	1126.2
脚钉	6.8级	M16X180									2.3	2.6	2.6	2.9	4.9	2.0	4.2	7.2	3.0	5.2	3.0	7.8	3.9	5.2	14.7	16.9	19.9	20.9	23.5	24.8	14.7	16.9	19.9	20.9	23.5	24.8
		M20X200									1.2	1.9	1.9	2.5	1.9	0.6	0.6	0.6	0.7	0.6	0.7			0.6	8.1	8.1	8.1	8.8	8.2	8.8	8.1	8.1	8.1	8.8	8.2	8.8
	8.8级	M24X240														1.8	1.8	1.8	2.4	1.8	2.4	3.6	1.8	1.8	1.8	1.8	1.8	4.2	6.0	6.0	1.8	1.8	1.8	4.2	6.0	6.0
		小 计									3.5	4.5	4.5	5.4	6.8	4.4	6.6	9.6	6.1	7.6	6.1	11.4	5.7	7.6	24.6	26.8	29.8	33.9	37.7	39.6	24.6	26.8	29.8	33.9	37.7	39.6
垫圈	Q235	-3(Ø17.5)	0.3	0.2			0.3	0.3								0.2	0.3	0.3		0.3		0.3		0.3	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.9
		-3(Ø22)									0.1														0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
		-4(Ø22)									0.1														0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
		小 计	0.3	0.2			0.3	0.3			0.2					0.2	0.3	0.3		0.3		0.3		0.3	0.9	1.0	1.0	1.0	1.0	1.0	1.0	1.1	1.1	1.1	1.1	1.1
座板式合计 (kg)			854.0	394.6	1473.9	1612.5	918.4	548.0	1456.9	2665.0	769.0	1564.9	1568.1	2308.2	4176.2	3750.3	5157.7	6748.9	2303.1	6219.6	2223.4	8387.1	3234.2	6886.2	16903.6	18311.0	19902.2	21676.0	23763.8	25497.1	18160.1	19567.5	21158.7	22932.5	25020.3	26753.6
插入式合计 (kg)			854.0	394.6	1473.9	1612.5	918.4	548.0	1456.9	2665.0	769.0	1564.9	1568.1	2308.2	4176.2	2949.5	4294.0	5971.8	2303.1	5348.6	2223.4	7639.4	3234.2	6001.2	16102.8	17447.3	19125.1	20805.0	23016.1	24612.1	17359.3	18703.8	20381.6	22061.5	24272.6	25868.6

螺栓增重表

塔型	330-KC22D-DJ (0-40度)						330-KC22D-DJ (40-90度)					
呼称高(m)	15	18	21	24	27	30	15	18	21	24	27	30
螺栓增重(kg)	176.7	183.4	187.3	202	195	206.5	180	186.7	190.6	205.3	198.3	209.8

甘肃电通电力工程设计咨询有限公司

甘肃庆阳能化电厂330千伏送出工程 工程 | 施工图 设计阶段

批准	曹晓青	
审核	王公阳	
校核	丁国鹏	
设计	王文军	
CAD制图	段博强	
比例		
日期	2026年7月	

330-KC22D-DJ转角塔

总图及材料汇总表

图号 330-KC22D-DJ-01(2/2)